

A Human-Aligned, Interpretable Similarity Metric for Time Series

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Abstract

This document records the proposed framing of the project and a critical assessment of its novelty and validation strategy. The central thesis is that the proposed distance — built from change-point segmentation, hand-crafted shape features, and a *merge-aware* segment-level dynamic time warping — is designed not to optimise a benchmark loss, but to *reproduce the human notion of similarity* between sequences. The empirical case rests on a human study over *adversarial* examples on which the proposed metric and standard baselines disagree. We summarise why this framing is strong, the main threats to the human study’s validity, the most closely related prior work (verified against the literature), and concrete recommendations before final results are produced.

1 The thesis

The goal is a **human-aligned similarity metric**: a distance that represents the human notion of similarity between two sequences, rather than one that optimises an arbitrary objective. The argument has two pillars:

1. **Construction.** The algorithm encodes the structural cues a human reasons about — it segments a sequence at change points, describes each segment with interpretable shape features (trend direction/strength, curvature, amplitude, length), and aligns segments with a merge-aware DTW that can collapse several segments on either side into one matched unit.
2. **Validation.** An empirical human study over *adversarial* pairs — examples where the proposed metric and the baselines (DTW, LCSS, Euclidean) rank similarity in opposite directions — in which human raters adjudicate. If humans consistently side with the proposed metric, the claim of human-alignment is supported by evidence external to the model.

Interpretability matters because humans make decisions for reasons they can articulate; a metric whose alignment is human-readable (a labelled segment-to-segment mapping) is the mechanism through which the “reasoning” can be inspected.

Motivating application. Comparing signals such as COVID-19 epidemic curves: locate where a similar trajectory occurred before, and use that precedent to inform the next decision. This is a query-by-example retrieval setting where *segment-level shape*, not point-level alignment, is what makes two curves “the same kind of thing”.

2 Why this framing is strong

- It converts the project’s main apparent weakness into a strength. The many hand-crafted, intuition-driven choices (shape features, trend labels, segment merging) would normally be attacked as arbitrary in an “accuracy” paper. Under the human-alignment thesis they are *defensible by design*: they encode the cues humans use.
- It moves the work out of the saturated “yet another DTW variant” competition (where the bar is beating dozens of methods on the UCR archive) into the less-occupied niche of *perceptually / human-aligned* similarity.
- Interpretability stops being a feature and becomes evidence for the thesis.

3 The human study is the paper

With this framing the empirical case rests almost entirely on the human experiment, so its methodology *is* the contribution. The predictable threats, and how to defuse them:

1. Cherry-picking / selection bias (the #1 threat). If adversarial examples are hand-selected where the method wins, the rebuttal writes itself: “you chose examples engineered to flatter your method.” *Mitigation*: make selection *mechanical and outcome-blind*. Define “the baselines disagree with us” by an automatic criterion, sample from that pool, and **freeze the set before observing which side the human took**. Present both directions (ours-high/theirs-low and ours-low/theirs-high); this symmetry is what buys credibility.

2. Ask humans the right question. Do *not* ask people for similarity scores or full rankings — these are noisy and unreliable. Use **triplets / odd-one-out**: “Is query Q more similar to A or to B ?” Each method predicts an answer; agreement with the human majority is the outcome. This is the standard robust design for perceptual-similarity studies and sidesteps most reliability complaints.

3. Measure inter-rater agreement — and worry if it is low. The premise is that “the human notion of similarity” is a real, shared target. If raters do not agree with *each other* (report Krippendorff’s α or similar), the target is ill-defined and the thesis weakens. **Check this early — it is an existential risk, not a detail.**

4. Avoid circularity. The metric is built from human intuition, then validated against humans. This is sound *only* if the validation set is genuinely held out and the metric is never tuned on the human judgments. Keep all weight tuning on a separate development set, sealed from the study data.

5. Keep a competence floor. If the only result is “we win on adversarial cases,” reviewers suspect a trade. Include a sanity check showing the metric is *competitive* (not necessarily SOTA) on ordinary cases / a standard benchmark, so the adversarial wins are not bought with ordinary-case losses.

Two claims, kept separate. Conflating these will hurt the paper:

- **Alignment:** the scores match human judgment — proven *only* by the human study.

- **Interpretability:** the method yields a human-readable segment alignment — proven by construction and examples.

Interpretability does not imply alignment. Treat them as two distinct contributions, each with its own evidence.

The application as motivation, not proof. The COVID retrieval use case is a strong story for the introduction, but proving it requires showing downstream *decision* quality, which is a different and harder paper. Safer: a qualitative query-by-example retrieval demo (“look how sensible the nearest neighbours are”) to illustrate the application, without resting validity on it.

4 Related work (verified)

The following items were checked against the literature in June 2026. The single most important one for positioning is DTW+S.

Human perception of time-series similarity. Gogolou et al. [1] ran controlled human studies on how well automatic measures (Euclidean, DTW) match human *perception* of similarity, and showed the visual representation (line chart vs. horizon graph vs. colorfield) changes that perception. This is the must-cite anchor. **Key differentiation:** they *measure* how existing metrics align with perception (descriptive); the present work *constructs* a new metric designed to be human-aligned and interpretable (constructive). It also primes reviewers to believe “DTW is already roughly perceptual”, so the study must beat DTW with statistical significance, not just on hand-picked cases.

Shape-based DTW variants (the methodological neighbours).

- **shapeDTW** [2]: encodes a local *shape descriptor* around every point and aligns descriptor sequences with DTW, yielding perceptually interpretable alignments; beats DTW on 64/84 UCR datasets. Differs from the present work in being *point-wise* (one descriptor per timestamp) rather than *segment-level*, and has *no human study*.
- **ESDTW** (extrema-based shape DTW) [5]: a more recent shape-DTW variant keyed on extrema — a useful comparison point for the segmentation idea.

The closest competitor — and the open gap. DTW+S (Srivastava, 2023) [3] is the nearest neighbour to this project: it builds an interpretable, “closeness-preserving” matrix in which each column is a *local trend*, then applies DTW — explicitly to match similar local trends at similar times, and explicitly motivated by **clustering epidemic / COVID curves** (surge → increase → peak → decrease). This overlaps the present project on three axes at once: shape/local-trend representation, interpretability, and the epidemic application. **Crucially, DTW+S contains no human study** — it is evaluated computationally (clustering / classification). This is the wedge: the present work’s human-alignment validation is exactly what the closest competitor does not do, and the segment-level *merge-aware* alignment (many-to-many segment collapsing with a correlation-gated merge cost) is a mechanism DTW+S does not have. **This paper must be cited, contrasted carefully, and ideally used as a baseline.**

Who is already building on DTW+S. Checking the papers that cite DTW+S confirms the line of work is active and epidemic-focused — and, tellingly, that citers *consume* it as a similarity primitive rather than scrutinising whether its alignments match human perception (which is precisely the present project’s wedge). The verified citers of DTW+S (the AAAI 2025 published version, “Aligning time-series by local trends”) are:

Citing paper	Year	$\approx$ cit.	How it uses DTW+S
TrendGNN: Towards Understanding of Epidemics, Beliefs, and Behaviors [4]	2025	< 5	Uses DTW+S to build the signal-similarity graph (retaining the top-5 trend-similar edges per signal) that feeds a GraphSAGE epidemic-forecasting GNN; reports DTW+S-constructed graphs beat lagged-correlation and random / fully-connected baselines at 2–4 week horizons. A pure <i>downstream use</i> of DTW+S, with no perceptual / human evaluation of the similarity itself.

Table 1: Verified citers of DTW+S.

Recent baselines and the broader “human-aligned similarity” trend.

- **Multiscale Dubuc** [6]: a 2024 new time-series similarity measure — a fresh baseline to include.
- **Seeing Eye to AI?** (CHI 2025) [7]: deep-feature metrics match human perception of *visualizations* (scatterplots), not time series. Shows the “does this metric see like a human?” question is active at top HCI venues — useful to cite as evidence the broader direction is timely, while noting it does not address time-series shape.
- Classic baseline surveys for similarity-measure comparison on classification [8] remain the reference for the competence-floor experiment.

Study-design methodology. Ordinal / triplet similarity indices [9] formalise interpretable alignment from *relative* (triplet / quadruplet) comparisons rather than scalar scores — a useful reference for justifying the triplet study design and for analysing the collected judgments.

5 Positioning at a glance

Method	Shape -based	Segment -level	Interpret.	Human study	Application focus
DTW / LCSS	no	no	low	no	general
shapeDTW [2]	yes	no (point)	med	no	UCR classification
DTW+S [3]	yes	trend cols	yes	no	epidemic curves
This work	yes	yes (merge-aware)	yes	yes	epidemic / decision support

Table 2: The combination of segment-level merge-aware alignment *and* a human-alignment study is the unoccupied cell.

6 Residual risks reviewers will probe

- **Hyperparameter sensitivity.** Many hand-set constants (feature set, hard-coded normalisation means/stds, `CHANGE_THRESHOLD`, `SEGMENT_PERCENTAGE`, domain-tuned weights). Ablations / sensitivity analysis are needed to show robustness.
- **Computational cost.** The merge-aware DTW searches over merge spans (d_i, d_j) and recomputes merged features inside the loop, giving roughly $O(n^2m^2)$ behaviour — a scalability question on long series. **[verify: profile and report; consider caching merged features / bounding the max merge span.]**
- **“DTW is already perceptual.”** Primed by [1]; answered only by a statistically significant win in the human study.

7 Recommended next steps

1. Lock the human-study design *before* producing final results: triplet / odd-one-out, outcome-blind mechanical selection of adversarial triplets, pre-registered set, inter-rater agreement reported.
2. Obtain and run **DTW+S** as a baseline; write the explicit contrast (segment merge mechanism + human validation).
3. Add a competence-floor benchmark (standard datasets) so adversarial wins are not read as trades.
4. Run hyperparameter ablations and a complexity/runtime analysis.
5. Keep the COVID retrieval as a qualitative motivating demo, not a validity claim.

References

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